

From Simple Chemistry to Complex Dynamics: A New Excitable Active Liquid System

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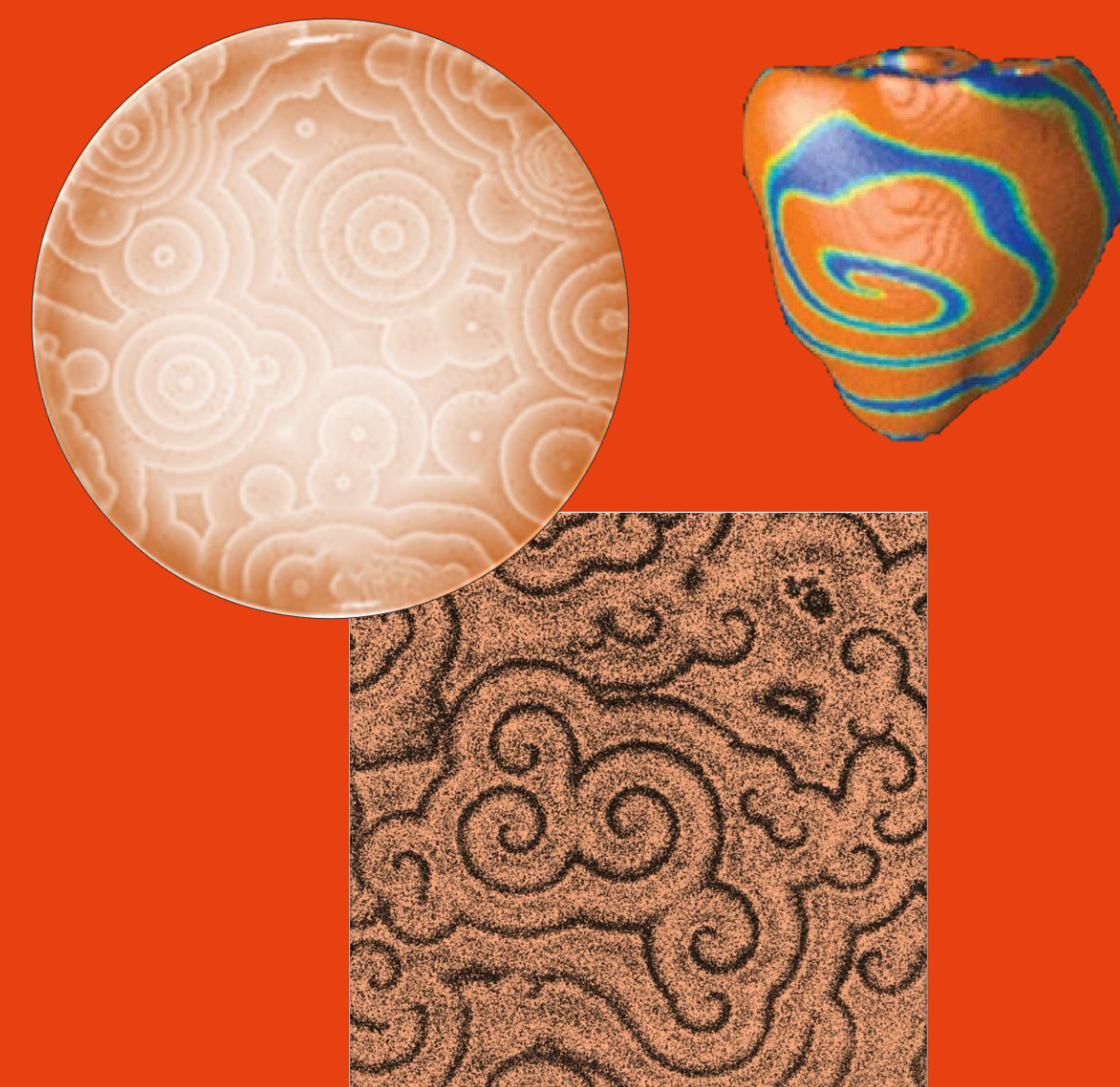
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INTRODUCTION

- Pattern formation, oscillations, and excitable media are widespread in nature, spanning physical, chemical, and biological systems.
- Classical examples include the Belousov–Zhabotinsky reaction, aggregation waves in *Dictyostelium discoideum*, and wave propagation in cardiac and neural tissue.



- Such systems are of particular interest to Artificial Life because they demonstrate how complex, adaptive behaviour can emerge from simple local interactions without central control.
- They provide experimentally accessible models for studying emergence, active matter, and the physicochemical principles underlying the transition from non-living to living systems.

MOTIVATION

- We discovered striking pulsatory behaviour of ethyl salicylate droplets on aqueous sodium decanoate solutions.
- Remarkably, the system is non-excitable when the solution is freshly prepared but develops robust quasi-periodic pulsations after aging.
- The aim of this work was to identify the physicochemical origin of this aging-induced transition and establish a reproducible experimental platform for studying the emergence of life-like dynamics in a minimal chemical system.

REFERENCES

- Čejková J. & Hauser M. (2026). *The Beauty of Patterns in Dictyostelium discoideum*, in *Understanding Marvelous Patterns in Complex Systems*, (Müller S.C. & Parisi J., Eds.), Springer.
- Čejková J. et al. (2017). *Artificial Life*, 23 (4), 528-549.

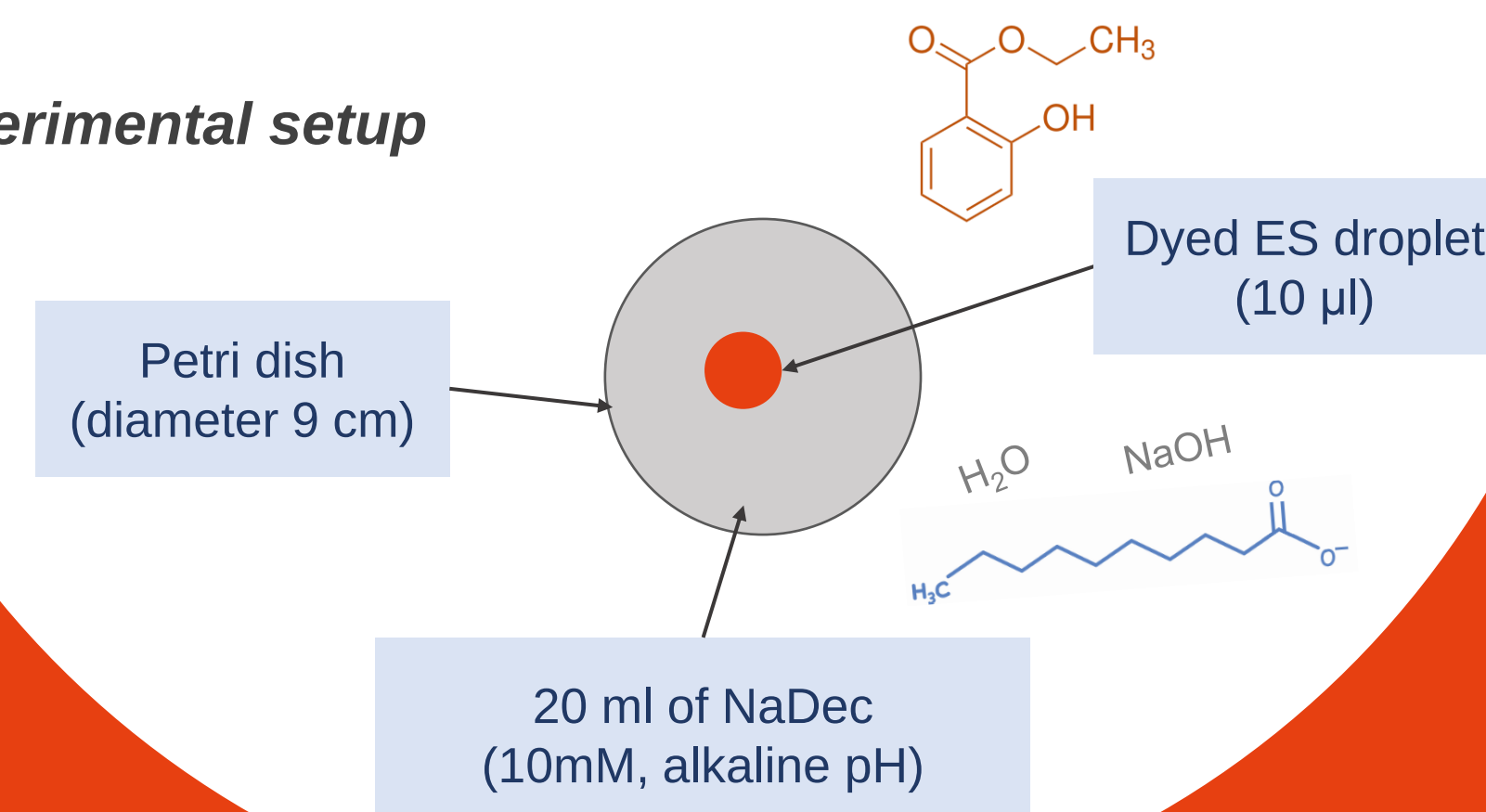
ACKNOWLEDGEMENT

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EXPERIMENTAL

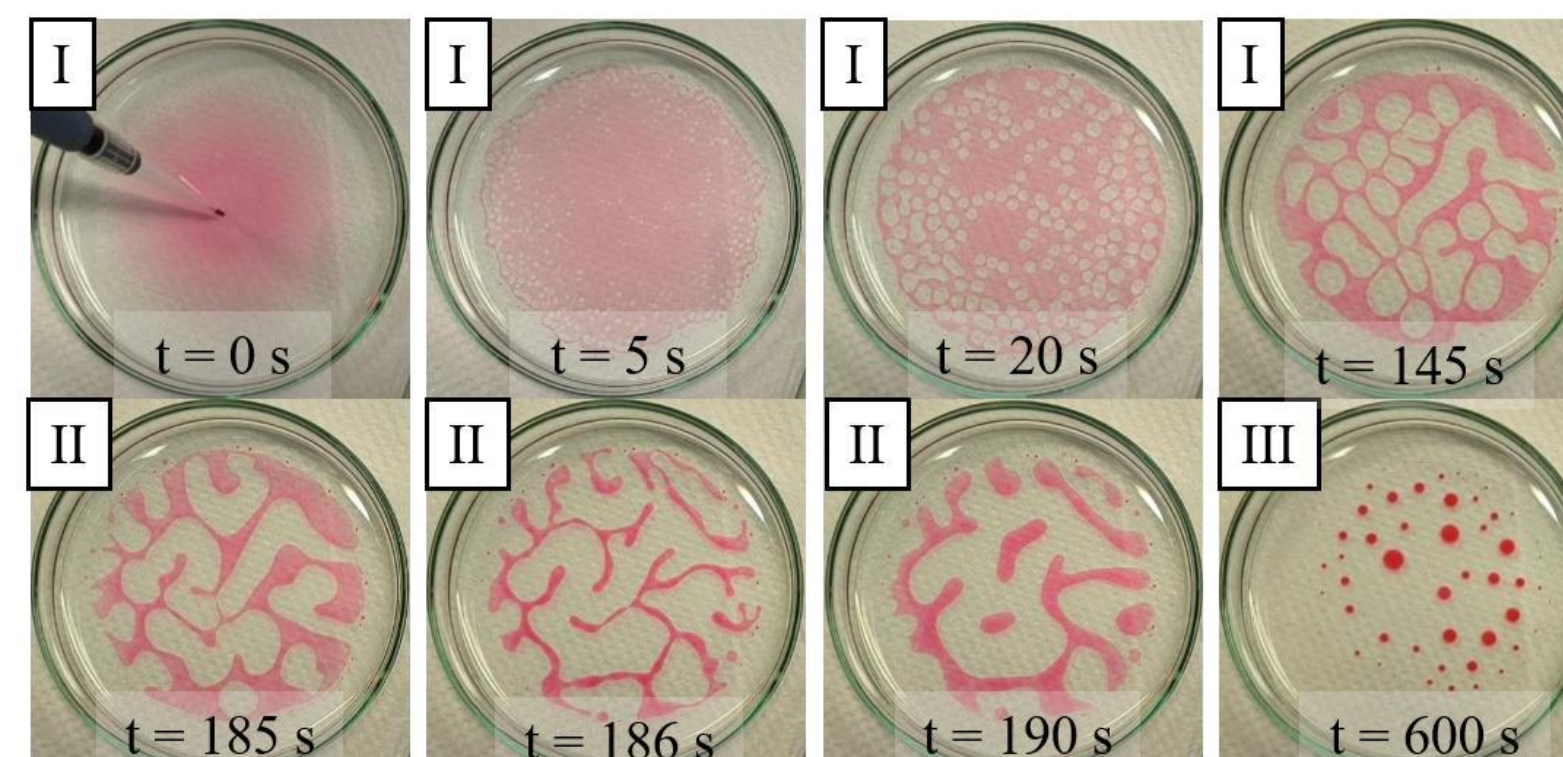
- Ethyl salicylate (ES) droplets were deposited onto aqueous sodium decanoate (NaDec) solutions.
- Experiments were performed using freshly prepared, aged, and chemically modified NaDec solutions.
- The effects of pH reduction, sodium silicate addition, and Na⁺ concentration were systematically investigated.
- A custom automated liquid-handling robot was designed and built for this work to ensure highly reproducible droplet deposition.

Experimental setup



RESULTS

- The system spontaneously evolved through three dynamical regimes:
 - (I) rapid spreading and hole nucleation
 - (II) self-organized network dynamics with quasi-periodic pulsations,
 - (III) fragmentation into isolated droplets.



- Pattern formation strongly depended on the choice of lipophilic dye.

CONCLUSIONS

- Chemical aging acts as a control parameter governing the emergence of life-like dynamics.
- The ES–NaDec system provides a reproducible experimental platform for studying self-organization, excitability, and active matter.
- Automated experiments combined with image analysis enable quantitative investigation of emergent behaviour.
- Identify the chemical origin of aging and its role in the emergence of excitability.
- Investigate how ionic composition and boundary conditions shape self-organization and pulse dynamics.
- Quantify physicochemical parameters to test the proposed mechanism.
- Explore the system as a minimal experimental model for studying active matter, protocell-like dynamics, and artificial life.

FUTURE WORK

The 2026 Conference on Artificial Life
Waterloo, Ontario, Canada
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RESULTS



Examples of experiments recorded
with the camera from above
(sped up from real time)

RESULTS

- Time-lapse image sequences were quantitatively analysed using a custom Python procedure.
- The ES equivalent diameter ($d_{eq} = 4S / O$), free NaDec area, and pulse frequency were quantified throughout the experiment.
- The equivalent diameter captured the morphological evolution of the ES structure and exhibited characteristic minima during pulsatory events.

